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Research article



Factors governing irrigation water productivity of rice in northwestern India: evidence from farmers' fields

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ABSTRACT

Northwestern India is a global groundwater depletion hotspot, with intensive agricultural water use causing an average groundwater table decline of ~0.4m per year. Farmers typically cultivate rice during the monsoon (kharif) season under puddled flooded conditions, driven by subsidized electricity and government assured minimum support prices for cereals crops. Research trials show mechanized dry-direct seeded rice (DSR) significantly reduces water use without compromising yields. However, evidence of irrigation water savings and improved irrigation water productivity (IWP) under farmers' practice remains limited. To investigate this, we installed 100 water flowmeters in farmers' fields, measuring seasonal water use and conducted group interviews to assess production practices and yields for DSR and puddled transplanted rice (PTR) fields. DSR showed higher rice yields and significantly greater IWP than PTR. Additionally, we found irrigation water application (IWA) in DSR to be only slightly lower than PTR, but remained 1.6–2 folds higher than research trials, indicating substantial potential for further reductions. Key drivers of water use differed substantially among DSR and PTR, with shifts in sowing dates offering further water-saving opportunities and number of wells in a village constraining irrigation. Importantly, farmers adopting DSR reported shifting from double cropping (rice-wheat) to triple cropping (rice-potato-wheat, rice-wheat-maize) to increase incomes, potentially offsetting water savings. Overall, DSR improves IWP and farmers can further reduce IWA through trainings on irrigation scheduling, freeing up electricity resources and government subsidies. However, technological solutions alone cannot resolve the groundwater crisis – system-wide policies balancing farmer incomes and water conservation will be essential.

1. Introduction

India is an agrarian nation where agriculture contributes over 18 % of the country's Gross Domestic Product. Furthermore, agriculture supports livelihoods of over 43 % of total population (World Bank, 2021). The green revolution in 1960s and 70s has been characterized by wide scale adoption of high yielding crop varieties, irrigation infrastructure development, use of machinery and chemical fertilizers. Through government support, the green revolution has transformed the hot and semi-arid northwestern India into its food basket despite only receiving an average 536 mm rainfall – most of which occurs during the relatively short monsoon months. As part of this intensification,

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Northwestern India has also undergone a large-scale shift in its cropping pattern, replacing sorghum (*Sorghum bicolor*) and pearl millet (*Cenchrus americanus*) with rice (*Oryza sativa*) during the kharif (monsoon) season, and substituting barley (*Hordeum vulgare*) and gram (*Cicer arietinum*) with wheat (*Triticum aestivum*) during the rabi (dry) season. This transformation has led to the establishment of the rice-wheat cropping system that still covers most of the agricultural landscape (Deb et al., 2024c; Dhanda et al., 2022). The state of Haryana noted a remarkable increase of 242 % and 305 % for rice and wheat yield, respectively in 2019 relative to 1960 (Department of Economic and Statistical Analysis, 2020). Notwithstanding its success in supporting India's food security, substantial unintended consequences include adverse environmental impacts such as rapidly declining groundwater tables and soil degradation (Aryal et al., 2020; Deb et al., 2024b; Joshi and Tyagi, 1991; Urfels et al., 2024). Rapidly falling groundwater levels are a critical concern for the government authorities, environmentalists and farmers.

Several factors drive the groundwater overexploitation in Haryana. Some of these include lack of groundwater governance systems (Chandra and Brozović, 2024) and heavily subsidized electricity in farms (Gupta, 2023), which were instrumental for the expansion of tubewells from ~5 million in 2000 to ~8 million in 2020 across Haryana (Jangra, 2023) that support food security but also unchecked groundwater pumping. For the corresponding duration, the average depth of groundwater table was observed to increase from 9.8 m below ground level (m bgl) to 20.2 m bgl (CGWB, 2023; Sitender et al., 2023). Satellite data gathered for over 10 years also validate this declining trend of groundwater table. For instance, Long et al. (2016) and Rodell et al. (2018) analyzed the Gravity Recovery and Climate Experiment (GRACE) dataset from 2002 to 2015 and reported an overall water storage loss of -19.2 ± 1.1 km^3/year across northwestern India. At the same time, groundwater pumping has a substantial impact on the strained energy system and government coffers that subsidize farmers' irrigation practices. Reversing the groundwater decline and stabilizing groundwater levels will require a substantial reduction in consumptive crop water use while reducing irrigation use could free up valuable energy and government resources.

Changes in management practices can reduce groundwater depletion and energy used for pumping but require careful consideration of how these interventions affect the water balance (Deb et al., 2025; Hellegers and van Halsema, 2021) and consideration of possible rebound effects (Grafton et al., 2018). These include adopting short duration rice varieties (Humphreys et al., 2010), alternate wetting and drying (Bouman and Tuong, 2001), replacing rice with other high value crops that consume less water (crop diversification/switching) (Brar et al., 2022; Dubey et al., 2024), and delayed transplanting (Rajput et al., 2024; Urfels et al., 2022). Another agronomic intervention that has high potential of saving irrigation water is direct seeded rice (DSR) (Kumar and Ladha, 2011). DSR is a rice cultivation method where rice seeds are sown directly by a seed drill, thus bypassing the conventional nursery preparation, transplanting and puddling of soil – which are water demanding activities. Field scale studies across the globe have underlined that irrigation water productivity (IWP) – also referred as the amount of grain produced per m^3 of irrigation input; can be elevated from 29 % to 53 % by adopting DSR over the conventional puddled transplanted rice (PTR) (Ishfaq et al., 2020; Jin et al., 2024; Yadav et al., 2011b, 2012). Additional benefits include lower methane emission, lower production cost and labor savings that contribute to sustainability outcomes (Kumar and Ladha, 2011). Importantly, DSR use has substantially increased since COVID-19, spurred by its labor saving benefits and is now promoted by State Governments in India and used by and estimated 15–20 % of farmers in Northwestern India. However, under both DSR and PTR establishments how much do farmers irrigate remains uninvestigated.

Enhancing IWP necessitates identifying the key governing variables that must be carefully evaluated to improve overall productivity. Since IWP depends on irrigation inputs and crop yield, it is influenced by various factors, including climatic conditions, biophysical characteristics, agronomic interventions, and socioeconomic variables (Descheemaeker et al., 2011; Li et al., 2016; Zwart and Bastiaanssen, 2004). Research trials on DSR establishment have reported crop yield to be at par with PTR (Alam et al., 2018; Devkota et al., 2020; Peramaiyan et al., 2023; Yadav et al., 2011a), the variables regulating the disparity in IWP among both establishments is Irrigation Water Application (IWA). Climatological variables such as solar radiation, precipitation and temperature are critical determinants of both crop water requirement and crop yield (Casolani et al., 2020; Deb et al., 2022; Sun et al., 2024). Similarly, bio-physical characteristics include soil and plant characteristics that drive the irrigation water requirement (Ali and Talukder, 2008; Thanh et al., 2023; Tripathi et al., 2021; Yi et al., 2022). Agronomic variables include crop management, weeding, and soil nutrient management that affect both crop yield and water requirement (Deb et al., 2014, 2024a). Socioeconomic factors, particularly, education and wealth have been shown to have a direct linkage with crop choice, irrigation infrastructure and how much to irrigate which subsequently control the IWP (Hatch et al., 2022; Liu et al., 2024; Tatis Diaz et al., 2022). Therefore, factors governing IWP require a comprehensive assessment of all these variables and how they affect IWP.

Based on the established background and problem, our study examines the factors influencing IWP under both DSR and PTR establishments, incorporating a comprehensive range of variables. The novelty to our study lies in two key aspects. First, we account for possible socioeconomic variables that are often overlooked in existing literature but are likely to impact IWP. Second, while previous studies primarily relied on quantitative analysis to determine the contribution of governing variables to IWP, our study adopts a mixed-method approach to identify and diagnose the key attributes controlling IWP. The specific objectives are a) quantification of plot scale IWP under DSR and PTR establishments across two districts in Haryana; b) identification of key attributes governing the IWP under the two establishments; c) develop recommendations for agronomic practices that can improve the IWP. The findings of our study will play pivotal role in devising policy interventions and farm-scale management strategies to enhance the IWP and contribute to sustainability by restoring the depleting groundwater table.

2. Study area and data

The study area comprises of two districts, Karnal and Panipat, in the state of Haryana in Northwestern India. The districts are characterized by semi-arid climate with average annual precipitation of 582 mm in Karnal and 680 mm in Panipat (Fig. 1a). Similarly, for the corresponding districts the average annual temperature is recorded as 28.3 °C and 24.2 °C, respectively. Both districts comprise of a flat topography ranging from 205 to 256 m amsl and are drained by the Yamuna River and its tributaries in the east. The soil texture of both districts is classified from sandy loam (north) to clayey loam (south) with higher soil moisture holding capacity in the southern region of the districts.

3. Methodology

To identify water use in rice production across the districts, we selected 100 farms (25 DSR and 25 PTR establishments in each district) where we deployed water flowmeters. To eliminate water losses, water flowmeters were installed at the outlets of the pumps which measured the volumetric IWA (Fig. 1b and c). We define IWP as the ratio of crop yield to the volume of irrigation water applied, calculating it according to Equation (1). The rice yield data was reported by the farmers for the years 2023 and 2024, while the IWA was measured for the corresponding rice seasons under both DSR and PTR establishments.

$$IWP = \frac{paddy\ yield}{irrigation\ water\ applied} \quad \text{(Equation 1)}$$

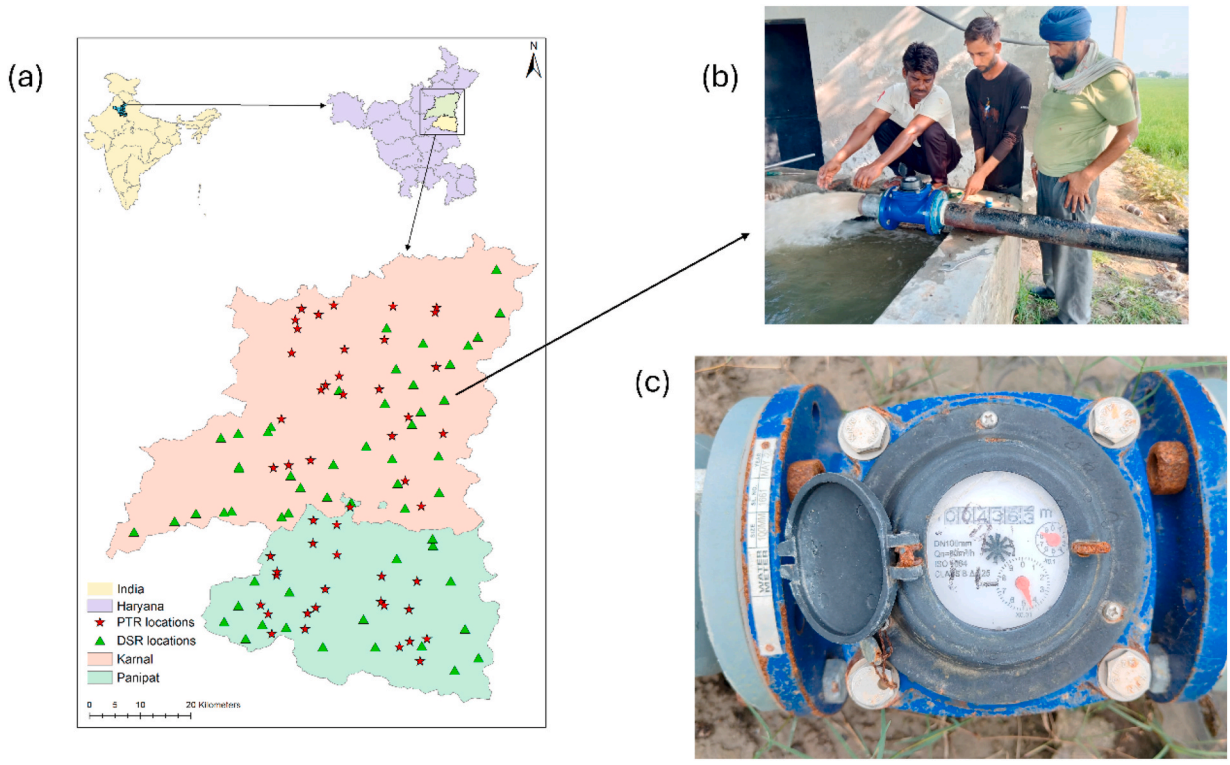


Fig. 1. (a) Location of study districts within India highlighting water flowmeter locations; (b) setup of the water flowmeter in the outlet of tubewell; (c) reading on a water flowmeter.

where IWP is in kg/m^3, paddy yield is in kg/ha, and irrigation water applied is in m^3/ha.

A comprehensive framework of the study is given in Fig. 2, where we employed a mixed method approach. First, we conducted a quantitative analysis to determine the contribution rate of farm intrinsic and extrinsic variables on IWP. Second, we performed a qualitative analysis based on four focus group discussions (FGDs) with farmers. Each FGD comprised of 10 farmers for each district and rice establishment method (i.e., PTR and DSR) – comprising a total of 40 farmers across Karnal and Panipat and DSR and PTR farmers. The discussions aimed at identifying the determinants of yield and IWA components of IWP. Additionally, we also identified the drivers of decision-making for irrigation frequency and duration.

3.1. *Quantitative analysis*

We assessed the contribution rates of the farm intrinsic and extrinsic variables on IWP by employing the transcendental logarithmic produc-

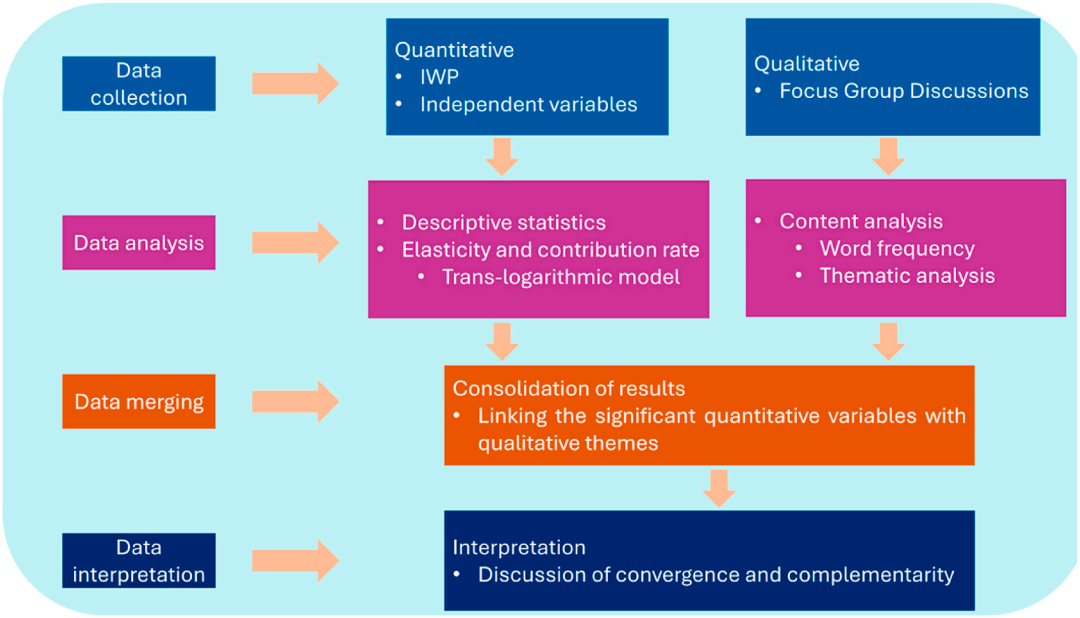


Fig. 2. Comprehensive flowchart of the work.

tion function or translog production function (TPF). The elementary model structure is given in Equation (2).

$$\ln WP = \beta_0 + \sum_{i=1}^{n} \beta_i \ln X_i + \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \beta_{ij} \ln X_i \ln X_j + \gamma_i Y_i + \delta_i Z_i \quad \text{(Equation 2)}$$

where WP is the output variable, X_i and X_j are the independent variables, Y_i and Z_i are the categorial variables, β_0 is the intercept, β_i are the first order effects, i.e., linear terms, β_{ij} are the second order interaction coefficients among the inputs, γ_i and δ_i are the coefficients of the dummy-encoded categorial variables. The coefficients represent the elasticity of the output variable to the independent variable. The independent variables collected in this study are date of sowing or transplanting; number of plots irrigated per tubewell - refers to the total number of plots irrigated by a tubewell; antecedent root zone soil moisture - refers to the average soil moisture (%) during post-wheat and pre-paddy seasons; well depth unit area defined as the fraction of the depth of submersible pump (m) and the farm area that is used to irrigate; pump size per unit area - indicates the ratio of rated horsepower (hp) of the pump used to the size (ha) of a given farm; diurnal temperature - refers to the mean value of the difference between the maximum and minimum temperature (in °C) averaged throughout the cropping season; sunshine hours; precipitation; family size; and education level of family head.

Equation (2) is a standard multivariate linear function and was solved by multiple linear regression method using the parcel data collected from the 100 farmers (50 each for PTR and DSR farmers). Partial least squares method was employed in estimating the model parameters. Using differential computation, the contribution rate of each factor was calculated as given in Equation (3) below.

$$E_i = \beta_i \frac{\Delta X_i / X}{\Delta Y / Y} \quad \text{(Equation 3)}$$

3.2. Qualitative analysis

Peoples' decisions on irrigation amount, duration and frequency are crucial in driving the IWP, we delved into these aspects of irrigation decision making under both DSR and PTR establishments. Since we conducted the discussions in *Hindi* language, prior to any analysis we transcribed the responses to English.

3.2.1. Content analysis

Content analysis employs rules of coding to condense a large number of words into smaller categories (Zaidman-Zait, 2014). Generally, it is used in social and health research and comprises both quantitative and qualitative approaches. The objectives can be accomplished using three approaches conventional, directed, and summative (Hsieh and Shannon, 2005). In this study we employed the third approach where a word frequency analysis was done followed by a thematic analysis.

3.2.2. Word frequency

We performed word frequency analysis on the farmers' responses during the FGDs. Since the responses generated a lot of textual data, we restricted the sample in the thematic analysis. Therefore, we conducted word frequency test in R programming using "tidyverse" (Wickham et al., 2019), "tidytext" (Silge and Robinson, 2016), and "tm" (Feinerer et al., 2008) packages. This resulted in 27 most frequently appearing words in DSR farmers' responses and 18 most frequently appearing words from the farmers' engaged with PTR establishment. Most farmers practicing DSR and PTR establishments used at least one of these words in their responses. The analysis was done separately for each crop establishment method.

3.2.3. Thematic analysis

To identify the themes, we elicited the verbatim discussions that included the 27 and 18 most frequent words for DSR and PTR, respectively and employed as statements (Austin et al., 2020; Butler-Kisber, 2010). These statements were meticulously understood, and the meanings were interpreted as signified by the recurrent occurrence of words participants quoted. Following this we classified and organized repeated ideas from the verbatim discussions. These were later aggregated and refined to eliminate the similarities and redundancy – leading to identification of the final themes (Saldaña, 2016). We derived these themes based on the word frequency and the interconnectivity among words. The themes were identified for both irrigation duration and frequency under DSR and PTR establishments.

3.3. Data merging and interpretation

We employed the convergent parallel approach (Tomasi et al., 2018) for data merging under the mixed-method approach. In this approach we explored the links between the most sensitive variables of IWP (from quantitative analysis) and the farmers' decisions to irrigate (from qualitative analysis). Following this the overall interpretation of the findings is discussed. A similar approach was also employed in past studies on water resources management (Babbitt et al., 2015; Urfels et al., 2020) and mental health (Austin et al., 2020).

4. Results and discussion

4.1. Crop yield

The crop yields under DSR and PTR for the years 2023 and 2024 are reported in Fig. 3. The results demonstrate that the yield ranges from 5.7 t/ha to 6.3 t/ha and 5.8 t/ha to 6.1 t/ha under DSR and PTR, respectively. Additionally, the farmers reported higher yield under DSR relative to PTR for the year 2024 (6.3 and 6.2 t/ha for DSR and correspondingly 6.1 and 6.0 t/ha for PTR at Karnal and Panipat, respectively). Furthermore, across both years PTR shows higher yield stability across the landscape, while DSR demonstrates yield variability. This is represented by the wider band of yield shown in Fig. 3. It is worth noting that higher concentration of crop yield in DSR relative to PTR establishment can be seen in the top quantile (Fig. 3a to d). Overall, the reported yield from both DSR and PTR establishments for the variety PB 1509 is comparable. Similar results of equivalent or slight lower rice yield were also reported by Ahmed et al. (2025) and Peramaiyan et al. (2023) across Bangladesh and Odisha in India, respectively.

4.2. Irrigation water productivity (IWP) from DSR and PTR establishments

The results of IWP across the two districts are shown in Fig. 4. The median values of IWP range from 0.21 kg/m^3 to 0.37 kg/m^3 and 0.16 kg/m^3 to 0.28 kg/m^3 for DSR and PTR respectively. This indicates that higher paddy yield can be attained with lower IWA for the PB 1509 variety. Additionally, for both 2023 and 2024, across both districts a bimodal distribution of IWP can be seen except for PTR in Panipat for the year 2024 where a gamma distribution can be seen. This indicates that majority of the recorded IWA are dispersed from the median values towards the low and high extremes. Overall, the results demonstrate that at a landscape scale higher IWP can be attained through DSR.

Although, DSR yields higher IWP across field measurements compared to PTR establishment, the magnitude is significantly lower when compared to IWP obtained at research trials. For instance, water management trials setup at Panipat under DSR during 2023 and 2024 resulted in an enhanced median IWP within the range of 105 %–228 % relative to the IWP obtained from farmers' fields across both districts. Similarly, in case of PTR establishment the enhancement can be seen within the range of 85 %–225 % relative to the IWP from farmers' fields. One of the primary causes of the of higher IWP from research trials is often attributed to the optimal management practices including precise

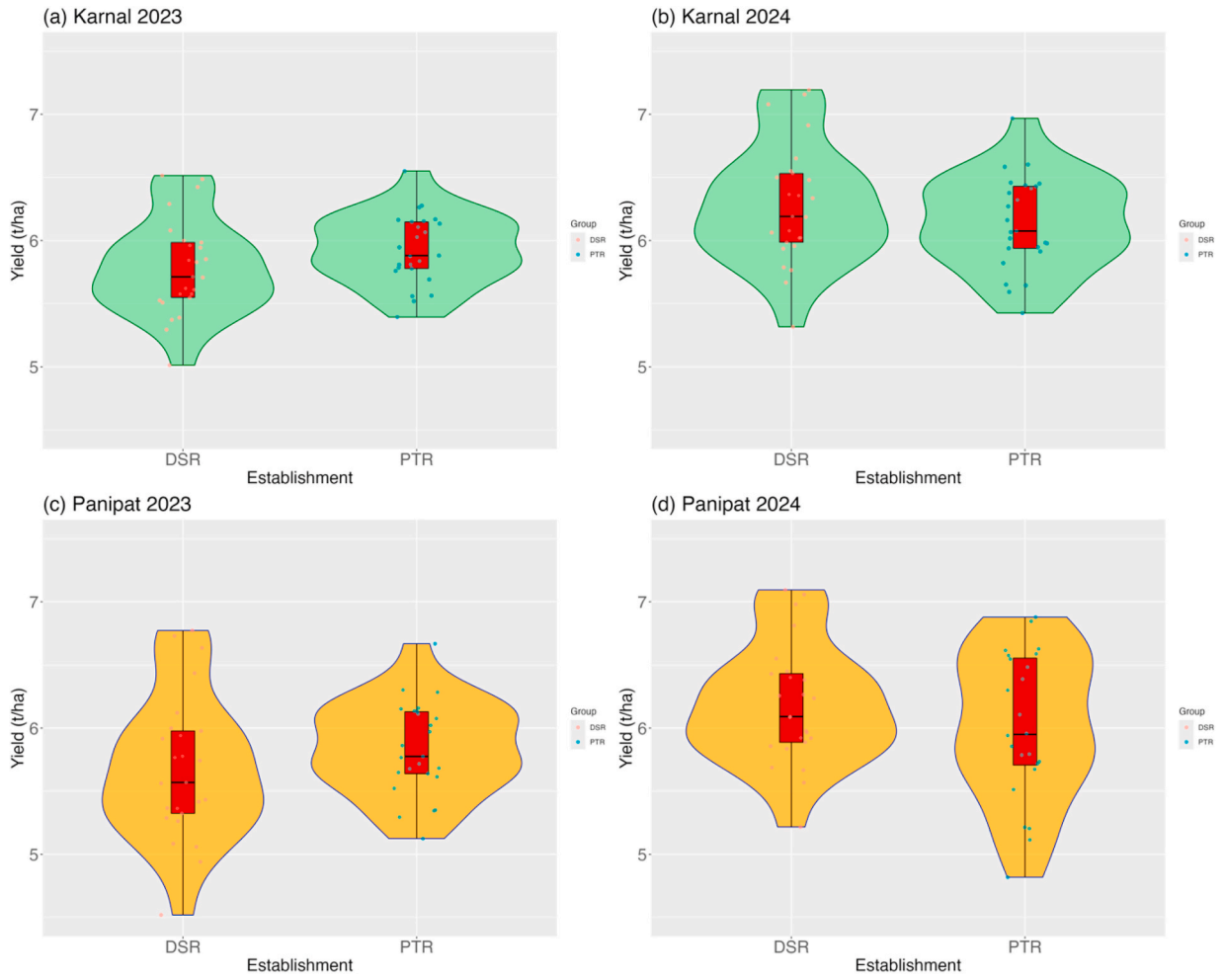


Fig. 3. Violin plot of paddy yield (t/ha) under DSR and PTR establishments at Karnal during (a) 2023, (b) 2024; and Panipat during (c) 2023, and (d) 2024.

water application, fertilizer and pest management (Bouman and Tuong, 2001). However, farmers' field conditions are generally subjected to variability in water availability, and application, and pest infestation which may reduce both crop yield and higher IWA – resulting to lower IWP (Lampayan et al., 2015). Overall, our results signify that there exists ample opportunity for improvement in IWP (by lowering the IWA) in farmers' fields, particularly in water scarce regions such as in Haryana. It is worth noting that for propriety reasons we are unable to report the results of field experiments conducted at Panipat.

4.3. Irrigation water application (IWA) from DSR and PTR establishments

In addition to IWP, we also analyzed IWA calculated for individual irrigation events and compared it against the IWA measured from research trials at Panipat. Results show consistently higher water application per irrigation for DSR than PTR across both districts. This is also found to be coherent in case of the research trials where 12.7 % higher water application is noted for DSR plots relative to PTR plots (Fig. 5a). Furthermore, we found that IWA ranges from 1.6 to 2 folds (~65 %–95 %) relative to the research trials. For PTR farms across both districts the average IWA per irrigation spans between 106.2 mm/ha to 134.1 mm/ha relative to 64.3 mm/ha, as in case of research trials. Likewise, for DSR farms, average IWA ranges from 123.5 mm/ha to 137.8 mm/ha, much higher than the research trials (72.5 mm/ha). Several interrelated factors are likely to contribute to this over-application. First, limited knowledge and adoption of irrigation scheduling often resulting in farmers relying on fixed calendars and visual cues, which tend to overestimate crop water needs (Kassam et al., 2019; Sarkar and Pal, 2020). Second, infrastructural constraints including low field application efficiency, seepage losses, and aging pump systems often prompt farmers to compensate by applying additional water (Perry et al., 2017). This is further exacerbated by the free access to electricity in farms (Gupta, 2023). Third, farmers' risk-averse behavior, shaped by past experiences with rainfall variability and uncertainty in water delivery, encourages higher than optimal irrigation as a buffer against perceived production risks (Koundouri et al., 2006; Pronti et al., 2024). Additionally, in case of DSR the absence of the clay layer in the root zone augments deep percolation resulting in a greater water application by the farmers (Yu et al., 2024). In contrast, for PTR, the clay layer aids in water ponding by eliminating deep percolation. Overall, farmers tend to apply significantly higher water than the crop water requirement as demonstrated by the research trials – showcasing the potential of higher water saving in the region.

In contrast to the above findings, we found a contrasting pattern of seasonal total IWA at both districts with lower volume of water applied in DSR compared to PTR. The total water savings under DSR establishment ranged from 22 % to 30 % across districts (Fig. 5b). Notably, we recorded the highest volume of IWA (29,500 m^3/ha) at Panipat under PTR, which was 203 % and 125 % higher than DSR and PTR farms, respectively, in the corresponding district. Similarly, at Karnal, the average seasonal total IWA was 22,223 m^3/ha and 17,286 m^3/ha under PTR and DSR, respectively. We recorded approximately seven additional irrigation events for PTR establishment compared to DSR establishment

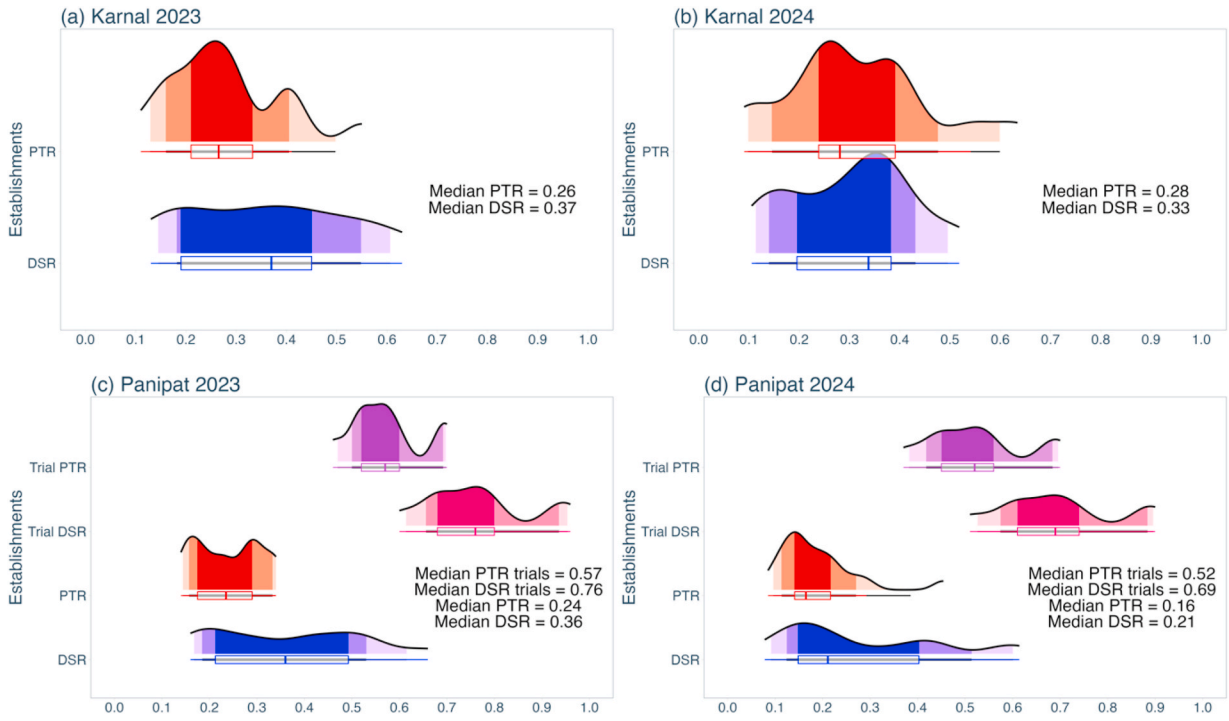


Fig. 4. Probability density plot of IWP in kg/m^3 under DSR and PTR establishments for Karnal during (a) 2023 and (b) 2024; and Panipat during (c) 2023 and (d) 2024, where an additional IWP is observed for trial plots under DSR and PTR establishments in Panipat for both 2023 and 2024. Note: The box in the boxplots shown in the base highlight the 25th to 75th percentile range of the IWP.

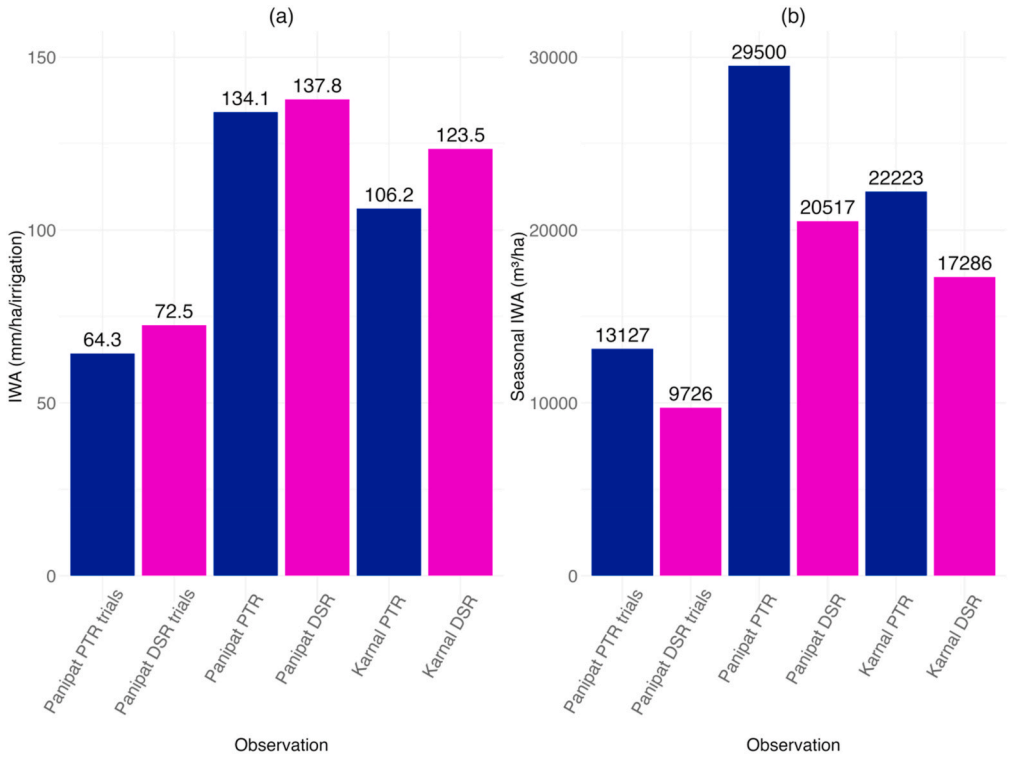


Fig. 5. (a) Average Irrigation Water Application (IWA) (in mm) measured for each irrigation per hectare and (b) Average total seasonal IWA in m^3 per hectare under DSR and PTR establishments across Panipat and Karnal districts. Note: first 2 bars in both plots represent the average IWA and seasonal total IWA for the PTR and DSR research trials set up at Panipat.

at both farmers' fields and research plots. The higher IWA in PTR can be primarily attributed to the higher evaporation losses during the initial growth stage, particularly when ambient temperatures are at their peak (Arora et al., 2006). Additionally, substantial water input is required for land preparation, particularly during the puddling process, which contributes 200–300 mm (~2000–3000 m^3) of IWA (Kar et al., 2018).

Overall, the higher seasonal total IWA for PTR is attributed to the higher number of irrigations as reflected by early maturity (compared to DSR), puddling requirement, constant requirement of ponding from tillering to panicle initiation, and recovering from transplanting shock (Mohammad et al., 2018).

4.4. Correlation between IWP and considered variables

The fundamental assumption of a TPF is that there exists a linear relationship between the independent and dependent variables. Therefore, to evaluate the relationship among the variables, we calculated Pearson correlation coefficients. It is worth mentioning that to avoid ordinality of the family size and education level of family head variables, we used only absolute values of the variables in the correlation estimation with IWP. This means while estimating the correlation with IWP, we considered the number of family members and the education level of household head, where education was represented numerically (e.g., 1, 2, 3 … …12 for high school levels, 13 for college first year, and so on).

The results of the correlation analysis are given in Table S1. We found date of sowing or transplanting, number of plots irrigated per tubewell, antecedent root zone soil moisture, and diurnal temperature variables have higher correlation with IWP relative to well depth unit area, pump size unit area, sunshine hours, family size, and education level of family head variables. Additionally, we observed precipitation to have insignificant correlation with IWP. This implies the farm intrinsic and management practices have a greater relationship with IWP compared to socio-economic variables. Furthermore, the negative correlations between IWP and number of plots irrigated per tubewell, antecedent root zone soil moisture, and pump size unit area demonstrate higher magnitude of IWP with lower values of the mentioned variables under both DSR and PTR establishments. It is worth noting that since precipitation does not have any significant relation with IWP, we did not include it in preceding analysis.

4.5. Quantitative analysis

4.5.1. Contribution rate

To determine the relationship between IWP and the considered variables, four generalized TPFs were developed for Karnal and Panipat under PTR and DSR establishments. The dataset used accounted for the spatial data collected for both 2022 and 2023 years. For model evaluation we employed Coefficient of determination (R^2), which was identified as 0.96, 0.87, 0.90, 0.88 for Karnal PTR, Karnal DSR, Panipat PTR, and Panipat DSR, respectively. These results demonstrate strong ability of the models in representing the relationships between IWP and the driving variables. Hence the variations in IWP can be effectively attributed to shifts in the driving variables.

The contribution rates (E in %) of the driving variables on IWP under the two establishments based on the TPF models are shown in Table S2. Basically, the contribution rate here demonstrates the impact of variations in driving factors (independent variables) on the growth of IWP. Our results suggest that IWP is either positively or negatively impacted by the considered variables under both PTR and DSR establishments. For instance, the contribution rates of the climatic variables, i.e., diurnal temperature and sunshine hours range from −3.2 to −15.8 and −2.6 to −5.7 across Karnal and Panipat under both establishments (Table S2). Similarly, the contribution rate of number of plots irrigated per tubewell to IWP is noted to be −22.8 (−22.4) and −15.0 (−13.5) for Karnal (Panipat) under PTR and DSR establishments, respectively. Notably, we found contrasting contribution rates of negative (−5.4 for Karnal and −2.6 for Panipat) and positive (19.9 for Karnal and 19.4 for Panipat) values for well depth unit area under PTR and DSR establishments, respectively. This implies that as tubewell depth increases, IWP is expected to rise significantly under DSR establishment (positive contribution rate) but decrease under PTR establishment (negative contribution rate). This is presumably because farmers generally adopt DSR when they experience lowering of groundwater table and tend to reduce the IWA, ultimately boosting the IWP (Rajput et al., 2024). Furthermore, the EDU variable is found to significantly contribute to IWP under both DSR and PTR establishments. Overall, the field conditions and farm associated variables such as date of sowing or transplanting, number of plots irrigated per tubewell, antecedent root zone soil moisture, well depth unit area, and sociological variable i.e., EDU significantly contribute to IWP.

The proportions of the contribution rates of the driving variables (E from Table S2) to IWP is presented in Fig. 6. It is to be noted that the proportions of the variables are scaled to “one” for demonstrating the relative contributions under the two establishments. We found that under DSR establishment, the relative contribution rates of field conditions (antecedent root zone soil moisture) and agronomic practice factor (date of sowing or transplanting) are highest compared to other variables at both districts. On the other hand, for PTR, the relative contribution rates of unit tubewell coverage of plots is found to highly control the IWP (number of plots irrigated per tubewell), followed by jointly education of family head (EDU) in case of both districts. A relatively high proportion of contribution is also found for climatic variable (diurnal temperature) at both districts – which is justifiable as it drives the crop water requirement. Overall, the findings imply that water/well availability; demography and climatic factors are critical in case of PTR. Similar results were also reported by Li et al. (2016) where IWP of paddy is found to be highly controlled by agronomic variables relative to climatic factors.

4.6. Qualitative analysis

To calculate IWP, both crop yield and IWA are crucial. Based on the open-ended question “What management practices do you implement to achieve the potential yield?” we found that 98 % of farmers follow the rice variety specific package of practices recommended by the district Krishi Vigyan Kendra (Farm Science Center), except for a higher fertilizer application rate – which is relatively uniform as per the participants. This leads to statistically insignificant difference in the crop yield, highlighting irrigation as the only sensitive variable. Therefore, the preceding analysis aims to diagnose the farmers’ irrigation frequency and duration decision making.

4.6.1. Word frequency

27 and 18 most iterated words of DSR and PTR practicing farmers in response to the question “How do you decide when and how much to irrigate?” are shown in Fig. 7. The most common words revolved around soil condition and power for DSR farmers and irrigation routine, inherited knowledge, and irrigation depth for PTR establishing farmers. Some of the other frequently appearing words are plant conditions and cropping cycle, and fixed interval and flooding for the corresponding classes of farmers. Additionally, we noted some redundant words which appeared among both classes such as fertilizer application. Overall, multiple variables were identified which are further discussed in the following section.

4.6.2. Themes

Based on word frequency, key themes were identified which are presented in Table 1. We found common themes to drive both irrigation frequency and duration decision making. These include irrigation characteristics among both DSR and PTR farmers with addition of power factor for the earlier establishment. Similarly, the responses from other participants can also be classified into plant characteristics, and weather conditions which drive the irrigation frequency under both establishments. Additionally, some of the responses on irrigation duration under the DSR establishment can also be classified into plot characteristics. Specifically, irrigation characteristics include responses including a fixed depth, uniform flooding, etc. To the question on irrigation frequency and duration, one of the DSR farmers responded:

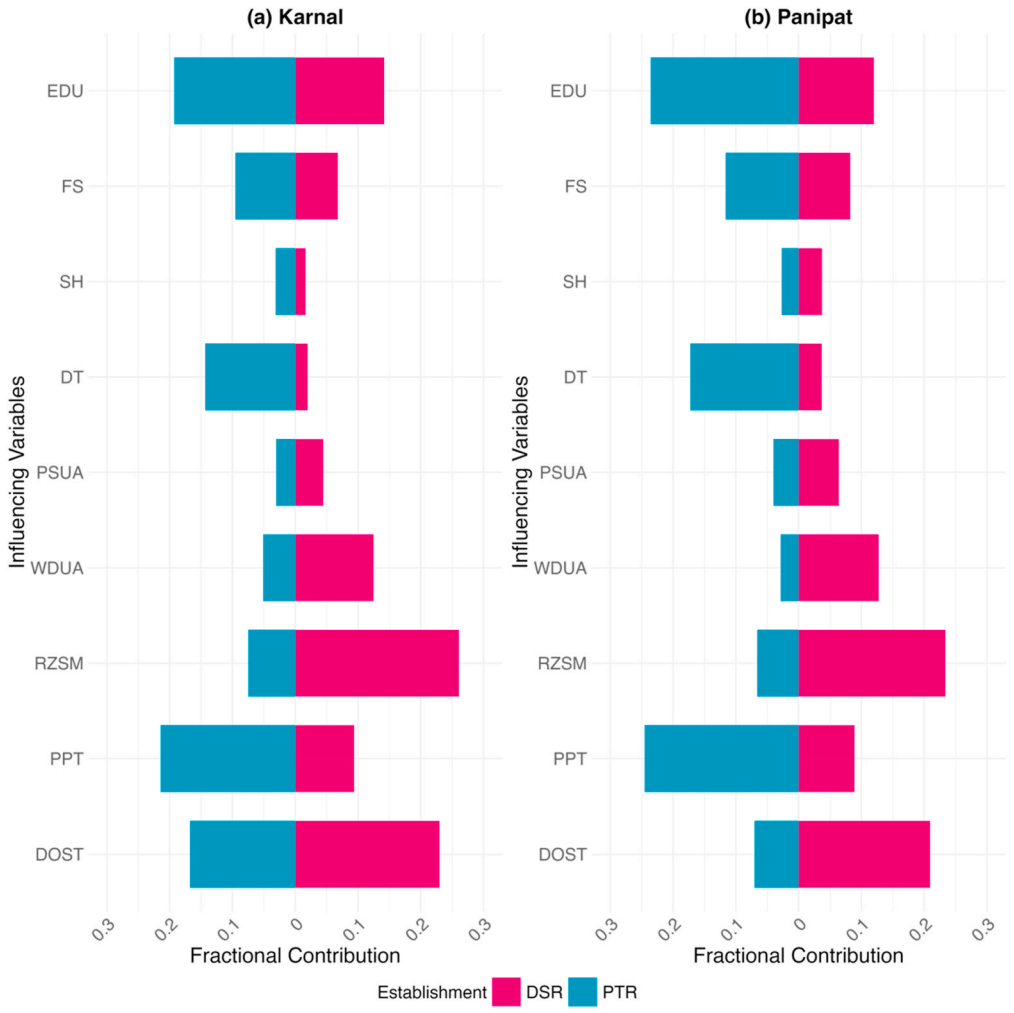


Fig. 6. Tornado diagram of fraction of the contribution rates of the influencing variables to IWP under DSR and PTR establishments. Note: DOST: date of sowing or transplanting, PPT: number of plots irrigated per tubewell, RZSM: antecedent root zone soil moisture, WDUA: well depth unit area, PSUA: pump size unit area, DT: diurnal temperature, SH: sunshine hours, FS: family size, and EDU: education level of family head.



Fig. 7. Word clouds of most frequent words in response to the open-ended question "How do you decide irrigation frequency and duration?" under (a) DSR and (b) PTR establishments. Note: larger (smaller) texts mean higher (lower) frequency of words.

Table 1
Identified themes driving irrigation frequency and duration under DSR and PTR establishments across Karnal and Panipat.

Establishment	Irrigation frequency	Irrigation duration
DSR	Soil factor	Plot characteristics
	Plant characteristics	Irrigation
	Weather	Power
	Irrigation	–
	Power	–
PTR	Irrigation	Irrigation
	Weather	Power
	Traditional knowledge	–

> The soil type in my village is slightly silty and I usually irrigate when the soil turns tan (dry) and generally irrigate until five centimeters.
>
> (Male, 53, Karnal)

Furthermore, another DSR farmer added "in my village there are frequent power cuts, therefore my irrigation schedule depends on the power availability. I usually irrigate two inches" (male, 45, Karnal). Following this a significant number of farmers also supported the statement. Similar responses were also observed in case of Panipat DSR farmers. Moreover, one farmer also responded:

> Generally, I walk on my plot to notice for my footprint and simultaneously if the crop leaves curl, I know it is the time to irrigate which is usually at 4–5-day interval. I generally irrigate until there is a uniform water distribution throughout the plot.
>
> (Male, 51, Panipat)

While we found the responses of the DSR farmers to incline towards soil moisture (such as "tan" and "moisture") and power availability, the PTR practicing farmers are noted to follow different criteria. There are reoccurring contextual themes as identified from the responses which navigate towards maintaining a consistent depth of water across the farm. The response from of one of the farmers was "My grandfather and father told me rice needs ponding water else there will be yield loss, therefore I irrigate everyday" (male, 47, Karnal). Additionally, another farmer (male, 55, Karnal) added "there is often low voltage in my village, and it takes roughly 2.5 h for each acre to irrigate, I need to irrigate whenever the voltage is higher to get my motor running. I usually maintain a depth of 15–16 cm". The responses highlight, farmers, regardless of soil conditions or crop water needs, tend to exert acquired knowledge, prioritizing maintenance of ponding throughout the crop duration, which leads to frequent irrigation and lower IWP. Additionally, under DSR conditions, the IWP is driven by the irrigation decision making attributed to soil moisture, crop characteristics and power availability.

4.7. Results consolidation and discussion

Our study compares yield and irrigation water use in practice between puddled transplanted and direct-seeded rice – a water saving technology. To achieve this we quantified IWP and the factors driving it under the conventional PTR and DSR establishments using data from the farmers' fields. The analysis highlights that higher IWP is achievable by adopting DSR compared to the conventional PTR practice. Although prior on-station experimental studies (research trials) have established similar findings (Alam et al., 2018; Yadav et al., 2011b, 2021), farmers practices remained poorly understood. Our results demonstrate that indeed a higher IWP within the range of 17.8 %–50.0 % can be obtained across the region for DSR farmers compared to PTR farmers – these are however less than in experimental station trials. The water saving in DSR is largely due to the accompanied adoptions of varities with shorter crop cycles and farmers irrigating with intermittent dry and wet cycles rather than constant flooding. Although we found some DSR farmers over-irrigate vis a vis crop requirements, much of the excess water is not lost and recharges aquifers through return flows. Additionally, reduced water use from DSR reduces on-farm energy consumption, with a scope to substantially lower the governments' energy subsidies costs.

One of the most critical variables identified in the quantitative analysis to control the IWP especially under DSR establishment is date of sowing or transplanting. Overall, a lower IWA is found for the farms where late sowing (early June) is done compared to the early sowing (beginning of May). This is because the late sowing coincides with the monsoon onset which generally meets the crop water requirement and reduce evapotranspiration. This finding aligns with other studies (Singh et al., 2019; Urfels et al., 2022) and corroborated during the FGDs with farmers expressing awareness of the high water use associated with early sowing. But the majority of the DSR farmers continue sowing early to achieve profit maximization through triple cropping. This is illustrated by statements such as follows:

> I think the soil drinks more water if I sow on early May as it is dry and hot. This increases the number of irrigations; however, I need to cultivate three crops in a year, and I have no choice than early sowing.
>
> (Male, 47, Karnal)

Additionally, we found only a fraction of DSR farmers (7 %) mentioning the keywords "sowing date" in their response to query on irrigation decision making. This implies that although farmers are conserving irrigation water by adopting DSR, there exists a higher potential of water saving, provided farmers adopt late sowing. However, economic incentives are currently not aligned with water conservation goals.

Broadly both districts exhibit similar patterns of influencing factors under DSR and PTR establishments, due to comparable cropping practices. Peer influence, especially from older farmers, plays a crucial role in shaping farmers' behaviors, as noted in studies by Johnson et al. (2023); Pathak et al. (2024) and Wang et al. (2024), which highlight the incorporation of elder farmers' experiences. The higher irrigation in PTR farms stems from the traditional approach to preventing water stress, supported by abundant groundwater until the late 20th century, when scarcity was not a concern (Kasana and Singh, 2017; Singh and Kasana, 2017). During the FGDs, we also discovered that power availability has a direct impact on IWP. Normally, farmers receive 8 h of electricity supply in the farms per day but occasionally the power is split into two shifts. If farmers irrigate during the first shift, they often re-irrigate the same plot in the second because they normally have a single pump inlet in the field and irrigation is carried out by topping field to field, leading to a higher water application and poorer IWP.

Overall, we discovered common patterns among the quantitative and qualitative analyses. For instance, antecedent root zone soil moisture is found to be critical across both districts in case of DSR. Similarly, descriptions of soil moisture specifying responses such as "moisture condition" and "footprint" are also detected in DSR farmers' responses during FGDs. Likewise, we found education level of family head is one of the most significant contributing variables to IWP in case of PTR, whereas "knowledge" is determined to be one of the most repeatedly appearing words in irrigation frequency decisions among the PTR farmers. It is worth mentioning here that "greater knowledge" does not refer to higher degree but more experience – either scientific or experiential knowledge. One of primary causes for the higher influence of education in improving IWP is likely because highly educated farmers are more aware of the environmental unsustainability by flooding irrigation, willing to adopt newer techniques and are more inclined to make weather-informed irrigation decisions (Jahangirpour and Zibaei, 2022; Xiuling et al., 2023; Zhang et al., 2022). Overall, the irrigation duration and frequency under PTR is found to be largely subjective of the electricity availability and farmer's fixed decision criteria for irrigation depth. However, DSR is driven by several factors including climate, soil

moisture and plant characteristics.

It is crucial to note that beyond the plot-level factors such as sowing date, soil moisture, well depth and density, and education level, broader institutional and behavioral drivers also shape the irrigation practices across these two districts. In particular, free electricity in farms encourages farmers to apply water beyond crop requirements (Fishman et al., 2015; Shah, 2009) facilitating groundwater pumping. Limited access to effective extension services and decision-support tool further compounds this tendency, as many farmers instead rely on traditional knowledge or peer practices that favor risk-averse strategies, including over-irrigation as a safeguard against rainfall variability and potential disruptions due to power cuts (Kaur et al., 2021). While a detailed quantitative analysis of these drivers is beyond the scope of the present study, acknowledging these factors helps contextualize why farmers' actual IWA is observed to be 1.6–2 folds higher than the experimental trials. Addressing these structural and behavioral influences through policy reform, targeted training, and improved advisory services could compliment technical interventions to promote more efficient water use in the region.

4.8. Farmers suggestions for potential strategies to decrease water use in rice production

During the FGDs some farmers mentioned that they experiment at their own farms. They are aware of the depleting groundwater table in the region and crop hyper-intensification (triple or more crops in a year) leading to straw burning and associated environmental concerns. A brief discussion on more sustainable cropping practice was also conducted leading to some points which can potentially boost IWP sustainably.

4.8.1. Splitting farms into smaller sub-plots

Literature reveals inconsistencies in reporting the average landholding size in Haryana, which typically ranges between 2.2 ha (5.4 ac) and 4.5 ha (11.3 ac) (Ajay et al., 2022). Regardless of the total landholding, farmers often divide their fields into 0.4 ha (1 ac) plots for easier management (Waha et al., 2020). During irrigation, pumps are operated continuously until each plot is completely watered. However, frequent power outages and split 8-h power supply into shifts often result in re-irrigation. To avoid this, some farmers advocate for dividing the plots into smaller units of 0.2 ha (0.5 ac) with added benefit of uniform water distribution across the plots. One of the farmers reported:

> The frequency of power cuts in my village has intensified over the last two years. This has led to reirrigation on the same block (sub-plot) as the water does not reach the tail end during the first split of power supply. Therefore, I have divided each block into 0.5 acres (~0.2 ha) and have made channels in the periphery of my plots for uninterrupted water flow to other sub-plots. Now the power cuts do not induce any re-irrigation of my sub-plots.
>
> (Male, 63, Panipat)

4.8.2. Tar-wattar DSR

Tar-wattar (delayed irrigation) DSR involves direct seeding under optimal soil moisture (or at field capacity), typically achieved by irrigating the field after land preparation and a few days prior to sowing (Yadav et al., 2013). Water is saved in this establishment since the subsequent irrigation is generally done 7–21 days after sowing, thus reducing 3–4 consecutive irrigations. This practice is generally not performed by the Haryanvi farmers, who prefer dry DSR over the tar-wattar DSR. In addition, weed pressure can be higher under Tar-wattar DSR and needs to be carefully managed. Studies have demonstrated that a delay in 21 days in the first irrigation post sowing had insignificant effect on rice yield, however, an overall reduction of 17.1 % in IWA (Kaur et al., 2024). This is because under conventional DSR practice, sowing is done in the first to second week of May when the crop water demand is at peak and with postponing the first irrigation up to 21 days can significantly reduce the overall IWA. Additionally, the succeeding water demand is met by the monsoon, reducing the irrigation requirements.

4.8.3. Avoid planting early during hot summer months

In addition to the above-mentioned strategies delayed sowing, as identified in this study, is likely the most important pathway for saving water and improve IWP, consistent with findings by Singh et al. (2019). Generally the farmers in Haryana follow the standardized package of practice from the Haryana Agricultural University to maximize the yield. This practice primarily focuses on optimizing inputs like fertilizers, herbicides and pesticides without specific guidance on IWA. As farmers over-irrigate due to subsidized electricity, our study emphasizes addressing the variable IWA component of IWP, particularly in different rice establishments like DSR and PTR.

4.8.4. Opportunities for materializing water saving through DSR adoption

The Haryana state government has been proactively promoting DSR by offering incentives of Rs. 4000 ($47) per acre, totaling Rs. 703 million (~$8.2 million) in 2023 (Deswal, 2024; Sharma, 2024). A total of 16 % of the rice area was already under DSR in 2024, but challenges like low germination, weed infestation, and limited seed drill availability constrained further expansion (Kaur and Kaur, 2017). Solutions to such problems exist including seed priming, herbicide application, and custom hiring centers (Kaur and Singh, 2017), highlighting the critical role of the private sector to enhance machinery availability, extension services and knowledge in scaling DSR adoption. Additionally, it is worth noting that securing single gradient slope (0.1 %) laser levelers can enhance IWP by 20–24 % relative to flat leveling, as they ensure even distribution of water in the root zone (Saleem et al., 2023; Winkler et al., 2018). Furthermore, it is critical to understand farmers' irrigation behavior, as applying the same irrigation practices of PTR for DSR can result in adverse outcomes. This is due to the absence of a plow pan in DSR, which increases infiltration that leads to less ponding and can thus lead to higher water usage and pumping (Humphreys et al., 2010).

The FGDs further revealed that farmers have identified a profitable strategy to maximize their earnings by utilizing DSR in conjunction with short-duration rice varieties. When questioned about their cropping intensity, 16 farmers representing 40 % of the total farmers, reported practicing a cropping intensity of 300 %. This cropping system is primarily adopted to maximize profit margins, as double cropping does not provide any significant financial benefits:

> "I generally take three crops including a short duration rice, then potato, followed by spring maize. The spring maize really gives me a good return as the minimum support price (MSP) is higher compared to wheat.
>
> (38, Male, Karnal)
>
> "I prefer growing three crops: rice, followed by potato, and then wheat. Since I rent the land for Rs. 70,000 (~$ 756) per acre per year, the rice-wheat cropping system does not even break even, so I must take three crops."
>
> (male, 47, Panipat)

We identified similar responses from the remaining 14 farmers, with 10 demonstrating a preference for the rice-potato-spring maize triple-cropping system. If this practice continues, it could pose significant sustainability challenges in two key aspects: a) spring maize, being a high-water demanding crop will necessitate increased IWA, potentially exacerbating the persistent water scarcity in the region; and b) the limited time interval between cropping seasons may lead to challenges in straw management, where crop residue burning could further accelerate, as it will become the most viable and easiest solution for farmers. These findings indicate that while DSR with short-duration varieties

enhances IWP, however, at cropping system level it may not necessarily reduce the IWA or contribute to mitigating groundwater depletion when implemented within a triple-cropping system involving spring maize on a large scale. Consequently, it is imperative to make substantial investments in identifying spring maize hotspots and guiding farmers towards diversifying into less water-intensive crops to ensure long-term sustainability.

With DSR being in existence for over 20 years and continuous government efforts to promote it, adoption has started to pick up especially given the increasing labor shortage and costs that were especially prevalent during the COVID-19 pandemic but continue to accelerate mechanization of agriculture. Even in areas where DSR is widely practiced, no significant improvement in the groundwater table has been observed over the past decade (Kumar et al., 2023). A likely reason for this is the irrigation decision-making process, which continues to rely on flooding. The study emphasizes that while DSR holds potential for improving IWP, irrigation decisions can be enhanced by incorporating weather forecasts. This could be achieved by improving local-scale predictions with longer lead times through ensemble-based probabilistic methods (Deb et al., 2022; Krivec et al., 2021; Tian et al., 2023). Moreover, adapting to frequent power outages is crucial for farmers, such as dividing fields into smaller units to mitigate power supply interruptions, which are exacerbated by rising power demand from population growth. Additionally, enforcing stricter policies on sowing dates is necessary, as many farmers prioritize higher profit margins and increased cropping intensity, leading to neglect of this crucial aspect.

5. Conclusions

IWP is a vital indicator with the potential to restore groundwater reserves in northwest India if effectively monitored. It is influenced by multiple drivers, and identifying these is essential for implementing sustainable policies and agronomic practices to enhance IWP. This is the first study in which we measured and reported IWA and IWP from farmers' fields at a landscape scale. We deployed 100 water flowmeters, evenly distributed between DSR and PTR farms across two rice-dominated districts in Haryana. We monitored seasonal IWA and calculated the IWP for 2023 and 2024 under both cultivation methods. Following this, we employed a mixed-method analysis incorporating variables such as biophysical characteristics, climate, farming practices, and demography, to identify key factors influencing IWP. Our findings suggest that, overall DSR farmers use less irrigation water than PTR farmers – but large variability exists within DSR farmers due to differences in antecedent soil moisture, and sowing dates. Additionally, power availability is crucial for affecting water use of all farmers. Furthermore, our results also suggest that the introduction of short-duration rice varieties with DSR is accompanied by a shift in farmers' cropping practice from the traditional rice-wheat system to rice-potato-wheat or rice-potato-spring maize systems. Triple cropping systems are likely to be more than offset any savings from the adoption of DSR, further exacerbating groundwater depletion. We conclude that water use can be reduced and IWP can be boosted through a combindation of (a) improved management practices (DSR) in the rice phase and (b) focusing on reducing overall system-level water use through adequate policy guidance and incentives. Addressing these factors can reduce irrigation pumping, ensuring the government's financial and policy efforts are effective and contribute to reaching the United Nations Sustainable Development Goals #2 (Zero Hunger) and #13 (Climate Action).

CRediT authorship contribution statement

Proloy Deb: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anton Urfels:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Virender Kumar:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Prakashan Chellattan Veettil:** Writing – review & editing, Supervision, Formal analysis. **Amit Kumar Srivastava:** Writing – review & editing, Methodology. **Swatantra Kumar Dubey:** Writing – review & editing, Visualization, Formal analysis. **Panneerselvam Peramaiyan:** Writing – review & editing, Formal analysis. **Sudhanshu Singh:** Writing – review & editing, Supervision, Resources. **Sandeep Rawal:** Writing – review & editing, Investigation, Data curation. **Satyendra Kumar:** Writing – review & editing, Formal analysis, Data curation. **Rajbir Garg:** Writing – review & editing, Data curation. **Baldev Raj Kamboj:** Writing – review & editing, Formal analysis. **Suresh Kumar:** Writing – review & editing, Data curation. **Buta Singh Dhillon:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2025.127428.

Data availability

Data will be made available on request.

References

Ahmed, S., Kumar, V., Zaman, A.U., Dewan, M.R., Khatun, A., Hossain, K., Singh, S., Timsina, J., Krupnik, T.J., 2025. Dry direct-seeded and broadcast rice: a profitable and climate-smart alternative to puddled transplanted aus rice in Bangladesh. Field Crops Res. 322, 109739. https://doi.org/10.1016/J.FCR.2025.109739.

Ajay, A., Craufurd, P., Kumar, V., Samaddar, A., Malik, R., Sharma, S., Ranjan, H., Singh, A., Paudel, G., Pundir, A.K., Poonia, S., Kumar, A., Kumar, P., Singh, D.K., Singh, M., Iftikar, W., Ignatius, M., Banik, N., Mohapatra, B., Sagwal, P., Yadav, A.K., Munshi, S., Panneerselvam, P., McDonald, A., 2022. Large survey dataset of rice production practices applied by farmers on their largest farm plot during 2018 in India. Data Brief 45, 108625. https://doi.org/10.1016/j.dib.2022.108625.

Alam, M.J., Humphreys, E., Sarkar, M.A.R., Yadav, S., 2018. Comparison of dry seeded and puddled transplanted rainy season rice on the high Ganges River floodplain of Bangladesh. Eur. J. Agron. 96, 120–130. https://doi.org/10.1016/j.eja.2018.03.006.

Ali, M.H., Talukder, M.S.U., 2008. Increasing water productivity in crop production—A synthesis. Agric. Water Manag. 95, 1201–1213. https://doi.org/10.1016/J.AGWAT.2008.06.008.

Arora, V.K., Gajri, P.R., Uppal, H.S., 2006. Puddling, irrigation, and transplanting-time effects on productivity of rice–wheat system on a sandy loam soil of Punjab, India. Soil Tillage Res. 85, 212–220. https://doi.org/10.1016/J.STILL.2005.01.015.

Aryal, J.P., Jat, M.L., Sapkota, T.B., Rahut, D.B., Rai, M., Jat, H.S., Sharma, P.C., Stirling, C., 2020. Learning adaptation to climate change from past climate extremes: evidence from recent climate extremes in Haryana, India. Int J Clim Chang Strateg Manag 12, 128–146. https://doi.org/10.1108/IJCCSM-09-2018-0065/FULL/PDF.

Austin, E.K., Rich, J.L., Kiem, A.S., Handley, T., Perkins, D., Kelly, B.J., 2020. Concerns about climate change among rural residents in Australia. J. Rural Stud. 75, 98–109. https://doi.org/10.1016/j.jrurstud.2020.01.010.

Babbitt, C.H., Burbach, M., Pennisi, L., 2015. A mixed-methods approach to assessing success in transitioning water management institutions: a case study of the Platte River Basin, Nebraska. Ecol. Soc. 20. https://doi.org/10.5751/ES-07367-200154.
Bouman, B.A.M., Tuong, T.P., 2001. Field water management to save water and increase its productivity in irrigated lowland rice. Agric. Water Manag. 49, 11–30. https://doi.org/10.1016/S0378-3774(00)00128-1.
Brar, A.S., Kaur, K., Sindhu, V.K., Tsolakis, N., Srai, J.S., 2022. Sustainable water use through multiple cropping systems and precision irrigation. J. Clean. Prod. 333, 130117. https://doi.org/10.1016/j.jclepro.2021.130117.
Butler-Kisber, L., 2010. Qualitative Inquiry: Thematic, Narrative and Arts-Informed Perspectives. SAGE, London.
Casolani, N., Cartone, A., Postiglione, P., Liberatore, L., 2020. Climate variability in agriculture and crop water requirement: spatial analysis of Italian provinces. J. Clean. Prod. 262, 121331. https://doi.org/10.1016/J.JCLEPRO.2020.121331.
CGWB, 2023. Dynamic groundwater resources of Haryana state. Faridabad.
Chandra, A., Brozović, N., 2024. Shifting dynamics and environmental implications of the irrigation pump market in India. Front. Environ. Sci. 12, 1398822. https://doi.org/10.3389/fenvs.2024.1398822.
Deb, P., Abbaszadeh, P., Moradkhani, H., 2022. An ensemble data assimilation approach to improve farm-scale actual evapotranspiration estimation. Agric. For. Meteorol. 321. https://doi.org/10.1016/j.agrformet.2022.108982.
Deb, P., Das, S., Patle, G.T., Elbeltagi, A., Yadav, S., 2024a. Development of pedotransfer functions for predicting saturated hydraulic conductivity in a Himalayan catchment: sikkim, India. Irrig. Drain. https://doi.org/10.1002/ird.2928.
Deb, P., Duong, T.A., Nguyen, D.M., 2014. Assessment of rice productivity under climate change for Ca Mau Province, Vietnam. In: Proceedings of the 19th IAHR-APD Congress 2014.
Deb, P., Kumar, V., Urfels, A., Lautze, J., Kamboj, B.R., Sharma, J.R., Yadav, S., 2025. Which machine learning algorithm is best suited for estimating reference evapotranspiration in Humid Subtropical climate? Clean 53, ee2023300441. https://doi.org/10.1002/CLEN.202300441.
Deb, P., Malakar, P., Bora, P.K., Dubey, S.K., 2024b. Univariate versus multivariate flood frequency analysis in tropical region: employing two classes of hydrological models. Clean. https://doi.org/10.1002/clen.202300351.
Deb, P., Urfels, A., Kumar, V., Dubey, S.K., Singh, J., 2024c. Water productivity in direct seeded rice and puddled transplanted rice: evidence from farmers' practices - CGIAR [WWW Document]. CGIAR Blogs. URL. https://www.cgiar.org/news-events/news/water-productivity-in-direct-seeded-rice-and-puddled-transplanted-rice-evidence-from-farmers-practices/.
Department of Economic and Statistical Analysis, 2020. Statistical Abstract of Haryana 2018-19. Chandigarh, India.
Descheemaeker, K., Amede, T., Haileslassie, A., Bossio, D., 2011. Analysis of gaps and possible interventions for improving water productivity in crop livestock systems of Ethiopia. Exp. Agric. 47, 21–38. https://doi.org/10.1017/S0014479710000797.
Deswal, D., 2024. 69% of claims of Haryana farmers for DSR incentive found fake - the Tribune [WWW Document]. The Tribune. URL. https://www.tribuneindia.com/news/haryana/69-of-claims-of-haryana-farmers-for-dsr-incentive-found-fake/.
Devkota, K.P., Yadav, S., Khanda, C.M., Beebout, S.J., Mohapatra, B.K., Singleton, G.R., Puskur, R., 2020. Assessing alternative crop establishment methods with a sustainability lens in rice production systems of Eastern India. J. Clean. Prod. 244, 118835. https://doi.org/10.1016/J.JCLEPRO.2019.118835.
Dhanda, S., Yadav, A., Yadav, D.B., Chauhan, B.S., 2022. Emerging issues and potential opportunities in the rice–wheat cropping system of North-Western India. Front. Plant Sci. 13, 832683. https://doi.org/10.3389/FPLS.2022.832683/BIBTEX.
Dubey, S.K., Veettil, P.C., Deb, P., Urfels, A., Kumar, V., 2024. Crop diversification for sustainable farming: what do farmers think? CGIAR [WWW Document]. CGIAR Blogs. URL. https://www.cgiar.org/news-events/news/crop-diversification-for-sustainable-farming-what-do-farmers-think.
Feinerer, I., Hornik, K., Meyer, D., 2008. Text mining infrastructure in R. J Stat Softw 25, 1–54. https://doi.org/10.18637/jss.v025.i05.
Fishman, R., Jain, M., Kishore, A., Ghosh, S., Singh, S., 2015. Groundwater depletion and farmer welfare in India. Sci. Adv. 1, e1500328.
Grafton, R.Q., Williams, J., Perry, C.J., Molle, F., Ringler, C., Steduto, P., Udall, B., Wheeler, S.A., Wang, Y., Garrick, D., Allen, R.G., 2018. The paradox of irrigation efficiency. Science 361, 748–750. https://doi.org/10.1126/SCIENCE.AAT9314/SUPPL_FILE/AAT9314_GRAFTON_SM.PDF.
Gupta, D., 2023. Free power, irrigation, and groundwater depletion: impact of farm electricity policy of Punjab, India. Agric. Econ. 54, 515–541. https://doi.org/10.1111/AGEC.12773.
Hatch, N.R., Daniel, D., Pande, S., 2022. Behavioral and socio-economic factors controlling irrigation adoption in Maharashtra, India. Hydrol. Sci. J. 67, 847–857. https://doi.org/10.1080/02626667.2022.2058877.
Hellegers, P., van Halsema, G., 2021. SDG indicator 6.4.1 "change in water use efficiency over time": methodological flaws and suggestions for improvement. Sci. Total Environ. 801, 149431. https://doi.org/10.1016/J.SCITOTENV.2021.149431.
Hsieh, H.-F., Shannon, S.E., 2005. Three approaches to qualitative content analysis. Qual. Health Res. 15, 1277–1288. https://doi.org/10.1177/1049732305276687.
Humphreys, E., Kukal, S.S., Christen, E.W., Hira, G.S., Singh, B., Yadav, S., Sharma, R.K., 2010. Halting the groundwater decline in North-West India—Which crop technologies will be winners? In: Bertsch, P.M., Phillips, R.L., Scow, K.M., Wildin, L. P., Boyer, J.S., Frey, K.J., Kamprath, E.J., Alexander, M. (Eds.), Advances in Agronomy. Elsevier, Amsterdam, Netherlands, pp. 155–217. https://doi.org/10.1016/B978-0-12-385040-9.00005-0.
Ishfaq, M., Akbar, N., Anjum, S.A., Anwar-Ul-Haq, M., 2020. Growth, yield and water productivity of dry direct seeded rice and transplanted aromatic rice under different irrigation management regimes. J. Integr. Agric. 19, 2656–2673.
Jahangirpour, D., Zibaei, M., 2022. Farmers' decision to adoption of modern irrigation systems under risk condition: application of stochastic efficiency with respect to a function approach. Front. water 4, 931694. https://doi.org/10.3389/FRWA.2022.931694/BIBTEX.
Jangra, P., 2023. Spatio-temporal variation in groundwater level due to tube well irrigation in Haryana state, India. NIU International J. Hum. Rights 7, 118–130.
Jin, Z., Harrison, M.T., Liu, K., Nie, L., 2024. Dry direct-seeded rice-wheat rotation system: lower water and carbon footprint and higher carbon production efficiency and net ecosystem economic benefits. Field Crops Res. 309, 109323. https://doi.org/10.1016/J.FCR.2024.109323.
Johnson, J.M., Becker, M., Dossou-Yovo, E.R., Saito, K., 2023. Farmers' perception and management of water scarcity in irrigated rice-based systems in dry climatic zones of West Africa. Agron. Sustain. Dev. 43, 1–18. https://doi.org/10.1007/S13593-023-00878-9/FIGURES/5.
Joshi, P.K., Tyagi, N.K., 1991. Sustainability of existing farming system in Punjab and Haryana: some issues on groundwater use. Indian J. Agric. Econ. 46, 412–421.
Kar, I., Yadav, S., Mishra, A., Behera, B., Khanda, C., Kumar, V., Kumar, A., 2018. Productivity trade-off with different water regimes and genotypes of rice under non-puddled conditions in Eastern India. Field Crops Res. 222, 218. https://doi.org/10.1016/J.FCR.2017.10.007.
Kasana, A., Singh, O., 2017. Groundwater irrigation economy of Haryana: a glimpse into spread, extent and issues. J. Rural Dev. 36, 531–556. https://doi.org/10.25175/jrd/2017/v36/i4/120624.
Kassam, S., Smith, M., Muñoz, G., 2019. Modernizing Irrigation: Improving Efficiency Through Management and Technology. Rome.
Kaur, D., Singh, A., Sindhu, V.K., 2024. Growth and productivity of direct-seeded basmati rice (Oryza sativa L.) as influenced by sowing dates and irrigation schedules in north-western India. Agric. Water Manag. 302, 108994. https://doi.org/10.1016/j.agwat.2024.108994.
Kaur, J., Singh, A., 2017. Direct seeded rice: prospects, problems/constraints and researchable issues in India. Curr. Agric. Res. J. 5, 13–32.
Kaur, K., Kaur, P., 2017. Reasons for adoption and non-adoption of direct seeded rice in Punjab. Indian j. ext. educ. 53, 157–159.
Kaur, R., Kaur, P., Sidhu, R.S., 2021. Farmers' awareness and perception towards efficient water management in Punjab and Haryana. Agric. Econ. Res. Rev. 34, 31–38.
Koundouri, P., Nauges, C., Tzouvelekas, V., 2006. Technology adoption under production uncertainty: theory and application to irrigation technology. Am. J. Agric. Econ. 88, 657–670. https://doi.org/10.1111/J.1467-8276.2006.00886.X.
Krivec, T., Kocijan, J., Perne, M., Grašic, B., Božnar, M.Z., Mlakar, P., 2021. Data-driven method for the improving forecasts of local weather dynamics. Eng. Appl. Artif. Intell. 105, 104423. https://doi.org/10.1016/J.ENGAPPAI.2021.104423.
Kumar, P., Ravi, Kumar, R., Mittal, S., 2023. Awareness status of direct seeded rice cultivation technology in Haryana state. IAHRW Int. J. Soil Sci. 11, 222–226.
Kumar, V., Ladha, J.K., 2011. Direct seeding of rice: recent developments and future research needs. In: Bertsch, P.M., Phillips, R.L., Scow, K.M., Wilding, L.P., Boyer, J. S., Frey, K.J., Kamprath, E.J., Alexander, M. (Eds.), Advances in Agronomy. Elsevier, Amsterdam, Netherlands, pp. 297–413. https://doi.org/10.1016/B978-0-12-387689-8.00001-1.
Lampayan, R.M., Rejesus, R.M., Singleton, G.R., Bouman, B.A.M., 2015. Adoption and economics of alternate wetting and drying water management for irrigated lowland rice. Field Crops Res. 170, 95–108. https://doi.org/10.1016/J.FCR.2014.10.013.
Li, X., Zhang, X., Niu, J., Tong, L., Kang, S., Du, T., Li, S., Ding, R., 2016. Irrigation water productivity is more influenced by agronomic practice factors than by climatic factors in Hexi Corridor, Northwest China. Sci. Rep. 6 (1 6), 1–10. https://doi.org/10.1038/srep37971.
Liu, Z., Qiu, H., Zhu, Y., Huangfu, W., Ye, B., Wei, Y., Tang, B., Kamp, U., 2024. Increasing irrigation-triggered landslide activity caused by intensive farming in deserts on three continents. Int. J. Appl. Earth Obs. Geoinf. 134, 104242. https://doi.org/10.1016/J.JAG.2024.104242.
Long, D., Chen, X., Scanlon, B.R., Wada, Y., Hong, Y., Singh, V.P., Chen, Y., Wang, C., Han, Z., Yang, W., 2016. Have GRACE satellites overestimated groundwater depletion in the Northwest India aquifer? Sci. Rep. 6, 24398.
Mohammad, A., Sudhishri, S., Das, T.K., Singh, M., Bhattacharyya, R., Dass, A., Khanna, M., Sharma, V.K., Dwivedi, N., Kumar, M., 2018. Water balance in direct-seeded rice under conservation agriculture in North-western Indo-Gangetic Plains of India. Irrig. Sci. 36, 381–393. https://doi.org/10.1007/S00271-018-0590-Z/METRICS.
Pathak, R., Magliocca, N.R., Kumar, M., Rathore, L., Moradkhani, H., 2024. Does the future look irrigated? Evaluating the likelihood of irrigation adoption within Alabama. Southeast. Geogr. https://doi.org/10.1353/sgo.0.a944197.
Peramaiyan, P., Srivastava, A.K., Kumar, Virender, Seelan, L.P., Banik, N.C., Khandai, S., Parida, N., Kumar, Vivek, Das, A., Pattnaik, S., Sarangi, D.R., Yeggina, P.K., Yadav, A., McDonald, A.J., Craufurd, P., Singh, S., Malik, R.K., 2023. Crop establishment and diversification strategies for intensification of rice-based cropping systems in rice-fallow areas in Odisha. Field Crops Res. 302, 109078. https://doi.org/10.1016/J.FCR.2023.109078.
Perry, C., Steduto, P., Karajeh, F., 2017. Does Improved Irrigation Technology save Water?.
Pronti, A., Auci, S., Berbel, J., 2024. Water conservation and saving technologies for irrigation. A structured literature review of econometric studies on the determinants of adoption. Agric. Water Manag. 299, 108838. https://doi.org/10.1016/J.AGWAT.2024.108838.
Rajput, J., Kushwaha, N.L., Sikka, A., Faiz Alam, M., Mahapatra, S., Sena, D.R., Singh, D. K., Sahoo, P.K., Mani, I., 2024. Water accounting of groundwater over exploited districts in Haryana and Punjab states to analyse impacts of water conservation

measures on water availability. Water Supply 24, 3093–3117. https://doi.org/10.2166/ws.2024.201.
Rodell, M., Famiglietti, J.S., Wiese, D.N., Reager, J.T., Beaudoing, H.K., Landerer, F.W., Lo, M.H., 2018. Emerging trends in global freshwater availability. Nature 557, 651–659. https://doi.org/10.1038/s41586-018-0123-1.
Saldaña, J., 2016. The Coding Manual for Qualitative Researchers, third ed. SAGE, Los Angeles.
Saleem, S., Hussain, I., Ullah, H., Iqbal, M.U., Aslam, U., Nasir, M., Khaliq, A., Masood, S. A., Rauf, H.A., Shah, A.N., Hussain, F., 2023. Gradient-based LASER land leveling increases the water use efficiency, growth, and yield of cotton crop under changing climate. Proc. Pakistan Acad. Sci.: B. Life and Environmental Sciences 60, 215–224. https://doi.org/10.53560/PPASB(60-2)813.
Sarkar, S., Pal, B., 2020. Farmers' perceptions and adoption of water-saving technologies in India. Agric. Water Manag. 237, 106148.
Shah, T., 2009. Taming the Anarchy: Groundwater Governance in South Asia, first ed., first ed. Routledge, Abingdon.
Sharma, P., 2024. How direct seeding method is growing on Haryana farmers - Hindustan times [WWW Document]. Hinduatan Times. URL. https://www.hindustantimes.com/cities/chandigarh-news/how-direct-seeding-method-is-growing-on-haryana-farmers-101707335557529.html.
Silge, J., Robinson, D., 2016. Tidytext: text mining and analysis using tidy data principles in R. J. Open Source Softw. 1, 37. https://doi.org/10.21105/joss.00037.
Singh, B., McDonald, A.J., Kumar, V., Poonia, S.P., Srivastava, A.K., Malik, R.K., 2019. Taking the climate risk out of transplanted and direct seeded rice: insights from dynamic simulation in Eastern India. Field Crops Res. 239, 92–103. https://doi.org/10.1016/j.fcr.2019.05.014.
Singh, O., Kasana, A., 2017. Management of groundwater resources for sustainable agriculture in Haryana. Trans. Inst. Indian Geogr. 39, 133–150.
Sitender, Chaudhary, B.S., Singh, A., Salita, Himani, 2023. Study of spatio-temporal variability of groundwater table in semi-arid Agrarian District of Haryana. J. Geol. Soc. India 99, 563–569. https://doi.org/10.1007/S12594-023-2345-6/METRICS.
Sun, J., Chen, L., Qi, P., Zhang, G., 2024. Agricultural irrigation water requirement and its response to climatic factors based on remote sensing and single crop coefficient method. Water Resour. Manag. 38, 5215–5233. https://doi.org/10.1007/S11269-024-03910-Z/METRICS.
Tatis Diaz, R., Pinto Osorio, D., Medina Hernández, E., Moreno Pallares, M., Canales, F. A., Corrales Paternina, A., Echeverría-González, A., 2022. Socioeconomic determinants that influence the agricultural practices of small farm families in northern Colombia. J. Saudi Soc. Agric. Sci. 21, 440–451. https://doi.org/10.1016/J.JSSAS.2021.12.001.
Thanh, P.N., Van, T. Le, Minh, T.T., Ngoc, T.H., Lohpaisankrit, W., Pham, Q.B., Gagnon, A.S., Deb, P., Pham, N.T., Anh, D.T., Dinh, V.N., 2023. Adapting to climate-change-induced drought stress to improve water management in Southeast Vietnam. Sustainability 15, 9021. https://doi.org/10.3390/SU15119021.
Tian, Y., Zhao, Y., Son, S.W., Luo, J.J., Oh, S.G., Wang, Y., 2023. A deep-learning ensemble method to detect atmospheric Rivers and its application to projected changes in precipitation regime. J. Geophys. Res. Atmos. 128, e2022JD037041. https://doi.org/10.1029/2022JD037041.
Tomasi, J., Warren, C., Kolodzey, L., Pinkney, S., Guerguerian, A.-M., Kirsch, R., Hubbert, J., Sperling, C., Sutton, P., Laussen, P., Trbovich, P., 2018. Convergent parallel mixed-methods study to understand information exchange in paediatric critical care and inform the development of safety-enhancing interventions: a protocol study. BMJ Open 8, e023691. https://doi.org/10.1136/bmjopen-2018-023691.
Tripathi, S.K., Sharma, B., Kumari, P., Deb, P., Ray, R., Denis, A.F., 2021. Evaluation of productivity, quality and economics of turmeric under different moisture regime and integrated nutrient management at Eastern indo-gangetic plains, India. Agric. Res. (Wash. D. C.) 10. https://doi.org/10.1007/s40003-020-00524-w.
Urfels, A., McDonald, A.J., Krupnik, T.J., van Oel, P.R., 2020. Drivers of groundwater utilization in water-limited rice production systems in Nepal. Water Int. 45, 39–59. https://doi.org/10.1080/02508060.2019.1708172.
Urfels, A., Mirzabaev, A., Bricx, S., Deb, P., Dewi, K., Evangelista, G., Flor, R.J., Harris, B., Jayasiri, M.M.J.G.C.N., Kishore, A., McDonald, A.J., Mondal, M., Quicho, E., Saito, K., Kumar, V., Laborte, A., 2024. Economics of sustainable irrigation in smallholder agriculture: implications for food security and climate action. Environ. Res. Lett. https://doi.org/10.1088/1748-9326/ad6201.
Urfels, A., Montes, C., Singh, B., van Halsema, G., Struik, P.C., Krupnik, T.J., McDonald, A.J., 2022. Climate adaptive rice planting strategies diverge across environmental gradients in the Indo-Gangetic plains. Environ. Res. Lett. 17, 124030. https://doi.org/10.1088/1748-9326/ACA5A2.
Waha, K., Dietrich, J.P., Portmann, F.T., Siebert, S., Thornton, P.K., Bondeau, A., Herrero, M., 2020. Multiple cropping systems of the world and the potential for increasing cropping intensity. Glob. Environ. Change 64, 102131. https://doi.org/10.1016/j.gloenvcha.2020.102131.
Wang, Y., Wang, Z., Zhao, M., Li, B., 2024. The influence of technology perceptions on farmers' water-saving irrigation technology adoption behavior in the North China Plain. Water Policy 26, 170–188. https://doi.org/10.2166/WP.2024.170/1376730/026020170.PDF.
Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T., Miller, E., Bache, S., Müller, K., Ooms, J., Robinson, D., Seidel, D., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., Yutani, H., 2019. Welcome to the tidyverse. J. Open Source Softw. 4, 1686. https://doi.org/10.21105/joss.01686.
Winkler, A.S., da Silva, J.T., Parfitt, J.M.B., Teixeira-Gandra, C.F.A., Conceço, G., Timm, L.C., 2018. Surface drainage in leveled land: implication of slope. Rev. Bras. Eng. Agrícola Ambient. 22, 77–82. https://doi.org/10.1590/1807-1929/AGRIAMBI.V22N2P77-82.
World Bank, 2021. World development indicators | DataBank [WWW Document]. URL. https://databank.worldbank.org/India-GDP/id/c557dff4.
Xiuling, D., Qian, L., Lipeng, L., Sarkar, A., 2023. The impact of technical training on farmers adopting water-saving irrigation technology: an empirical evidence from China. Agriculture 13. https://doi.org/10.3390/AGRICULTURE13050956. Page 956 13, 956.
Yadav, D.B., Yadav, A., Vats, A.K., Gill, G., Malik, R.K., 2021. Direct seeded rice in sequence with zero-tillage wheat in north-western India: addressing system-based sustainability issues. SN Appl. Sci. 3, 844. https://doi.org/10.1007/s42452-021-04827-7.
Yadav, S., Gill, G., Humphreys, E., Kukal, S.S., Walia, U.S., 2011a. Effect of water management on dry seeded and puddled transplanted rice. Part 1: crop performance. Field Crops Res. 120, 112–122. https://doi.org/10.1016/J.FCR.2010.09.002.
Yadav, S., Humphreys, E., Kukal, S.S., Gill, G., Rangarajan, R., 2011b. Effect of water management on dry seeded and puddled transplanted rice. Field Crops Res. 120, 123–132. https://doi.org/10.1016/j.fcr.2010.09.003.
Yadav, S., Humphreys, E., Li, T., Gill, G., Kukal, S.S., 2012. Evaluation of tradeoffs in land and water productivity of dry seeded rice as affected by irrigation schedule. Field Crops Res. 128, 180–190. https://doi.org/10.1016/j.fcr.2012.01.005.
Yadav, S., Malik, R.K., Humphreys, E., Kumar, V., Singh, S.S., Chauhan, B.S., Kamboj, B. R., Gathala, M., Jat, M.L., McDonald, A.J., Laik, R., 2013. Guidelines for Dry Seeded Rice (DSR) in the Eastern Gangetic Plains of India.
Yi, J., Li, H., Zhao, Y., Shao, M., Zhang, H., Liu, M., 2022. Assessing soil water balance to optimize irrigation schedules of flood-irrigated maize fields with different cultivation histories in the arid region. Agric. Water Manag. 265, 107543. https://doi.org/10.1016/J.AGWAT.2022.107543.
Yu, Q., Dai, Y., Wei, J., Wang, J., Liao, B., Cui, Y., 2024. Rice yield and water productivity in response to water-saving irrigation practices in China: a meta-analysis. Agric. Water Manag. 302, 109006. https://doi.org/10.1016/J.AGWAT.2024.109006.
Zaidman-Zait, A., 2014. Content analysis. In: Michalos, A.C. (Ed.), Encyclopedia of Quality of Life and Well-Being Research. Springer, Netherlands, Dordrecht, pp. 1258–1261. https://doi.org/10.1007/978-94-007-0753-5_552.
Zhang, K., Li, Y., Yu, Z., Yang, T., Xu, J., Chao, L., Ni, J., Wang, L., Gao, Y., Hu, Y., Lin, Z., 2022. Xin'anjiang nested experimental watershed (XAJ-NEW) for understanding multiscale water cycle: scientific objectives and experimental design. Engineering 18, 207–217. https://doi.org/10.1016/J.ENG.2021.08.026.
Zwart, S.J., Bastiaanssen, W.G.M., 2004. Review of measured crop water productivity values for irrigated wheat, rice, cotton and maize. Agric. Water Manag. 69, 115–133. https://doi.org/10.1016/j.agwat.2004.04.007.